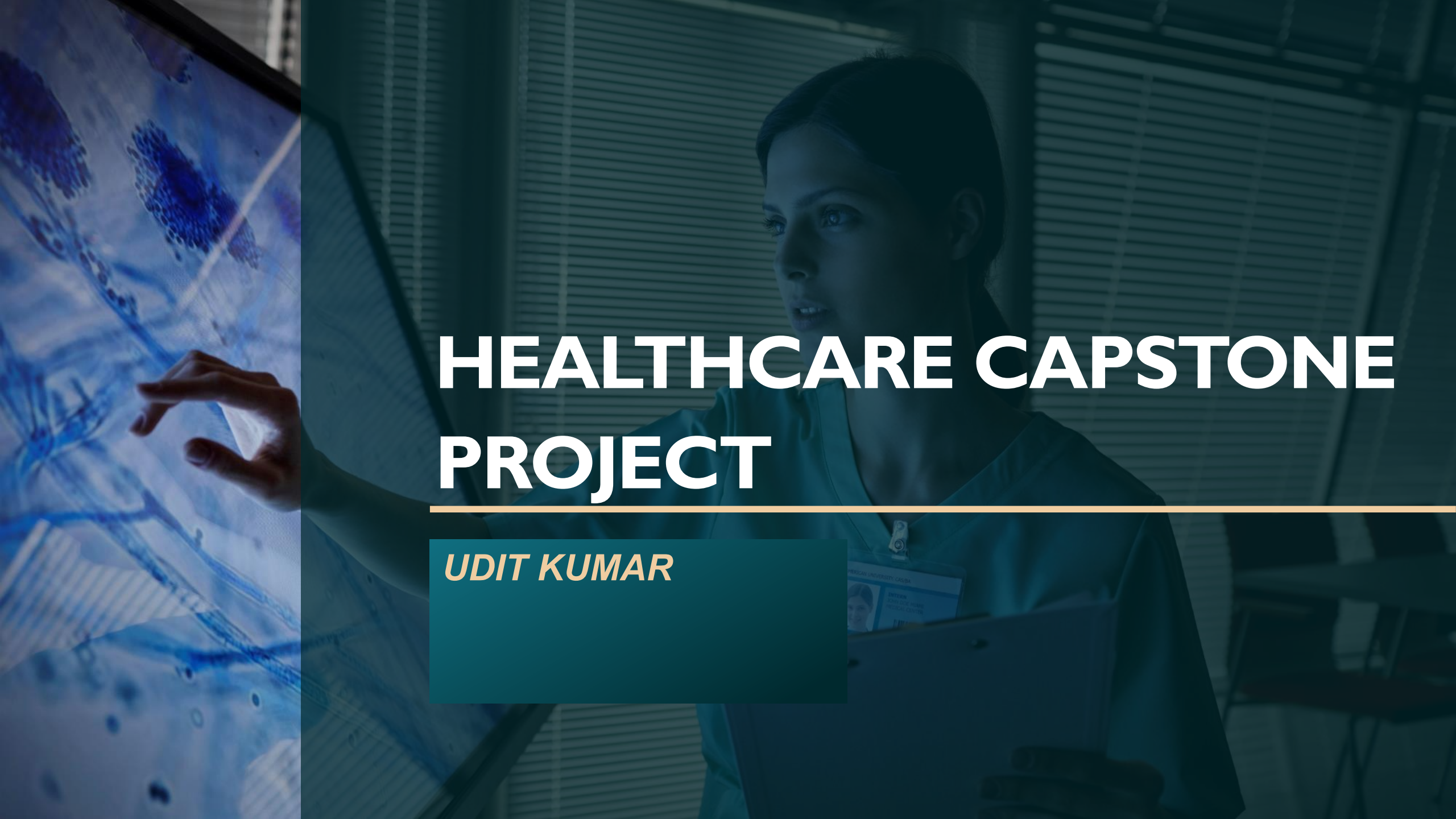

HEALTHCARE CAPSTONE PROJECT

UDIT KUMAR

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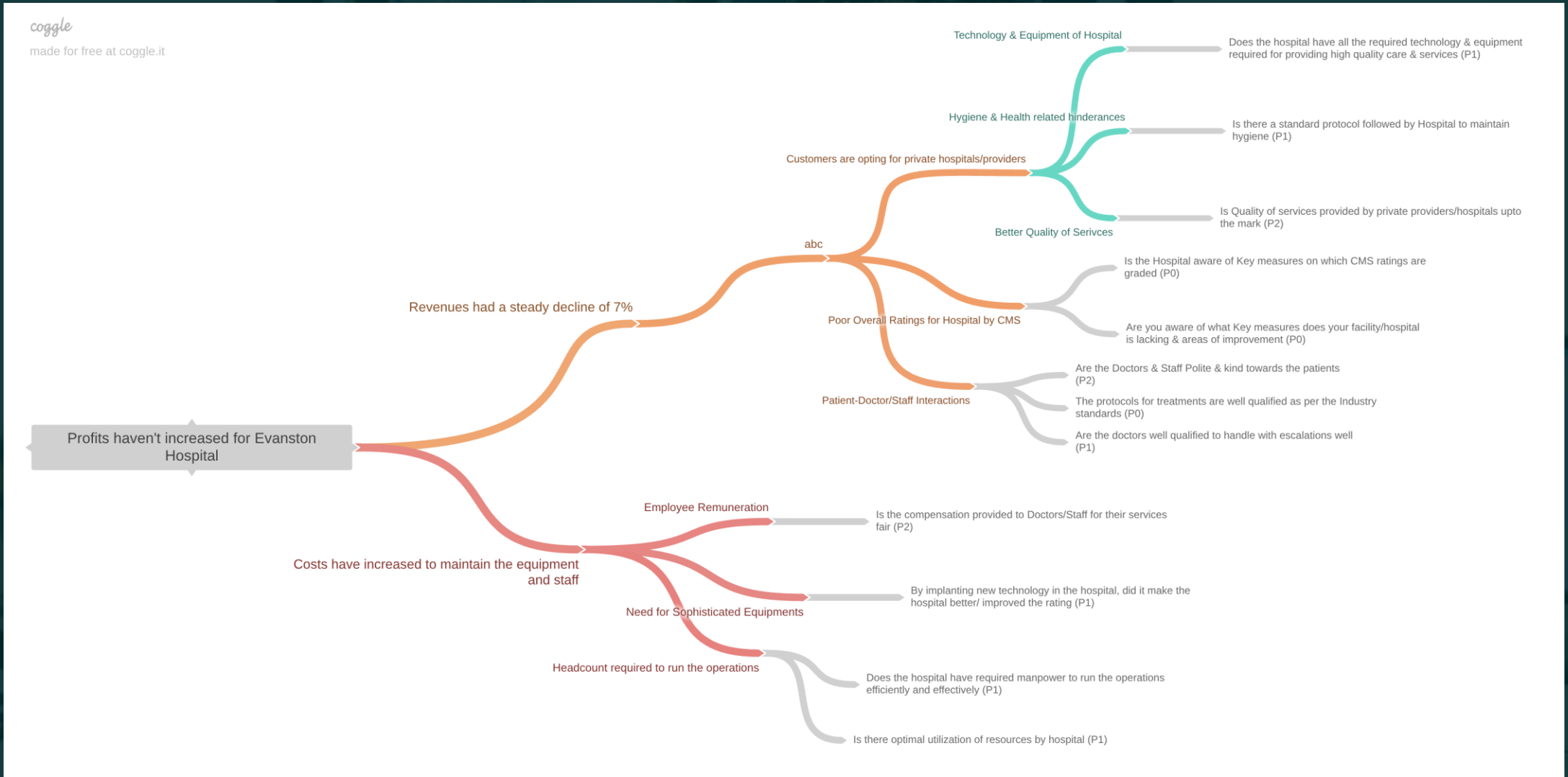
PROBLEM STATEMENT

Evanston Hospital, an acute-care facility, has maintained a stagnant overall rating of **3 out of 5 for the past five years**, resulting in declining revenue and reduced patient preference. Despite substantial investments to improve clinical quality and patient services, the hospital has not seen any meaningful rise in its performance score. The lack of progress indicates deeper underlying issues that have not been identified or addressed. A detailed analysis is required to understand the root causes behind the low ratings and determine which factors most strongly influence the hospital's overall performance.

Business Objectives

- **Uncover the main factors responsible for Evanston Hospital's sustained low ratings.**
- **Perform detailed analytical studies** to understand which indicators and performance measures most influence hospital ratings.
- **Develop predictive insights** to identify the strongest contributors to the hospital's rating, with an emphasis on Evanston Hospital's unique trends.
- **Highlight key improvement opportunities** among the top impact areas to help the hospital make targeted enhancements and raise its overall rating.

ROOT CAUSE ANALYSIS USING ISSUE TREE FRAMEWORK



HOSPITAL COMPARE & DATASET SELECTION

Hospital Compare is a patient-centric platform that provides insights into how well hospitals deliver necessary and quality care.

To identify the relevant datasets for our analysis, we referred to the document **"Overall Hospital Quality Star Ratings on Hospital Compare Methodology Report."**

This report outlines **62 performance measures**, categorized into **7 measure groups**, all of which are used by CMS to compute the hospital's overall Star Rating.

These groups and measures guided our dataset selection process.

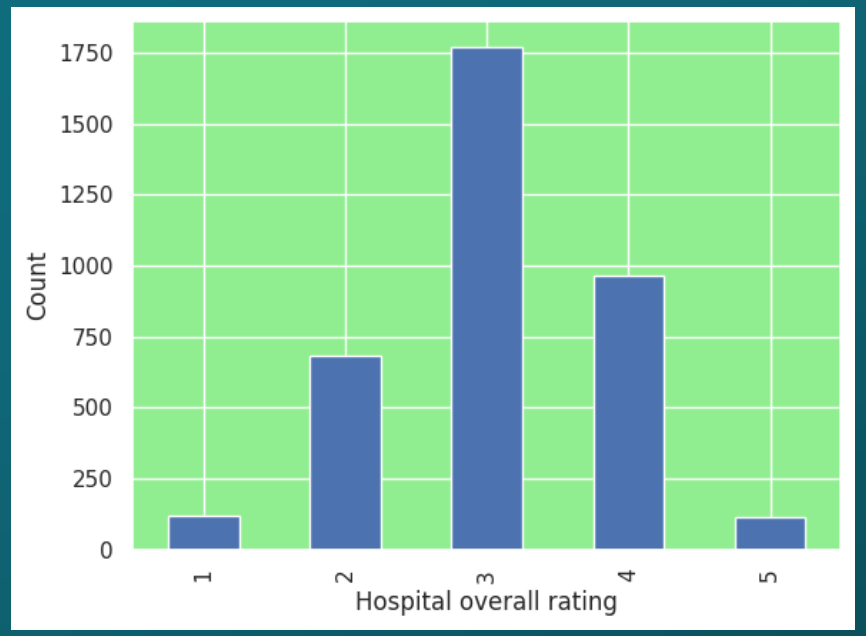
Dataset:
1) Mortality
2) Readmission
3) Safety of Care
4) Timeliness of Care
5) Effectiveness of Care
6) Patient Experience
7) Outpatient Imaging Efficiency

PROJECT WORKFLOW & CMS STAR RATING – SUMMARY

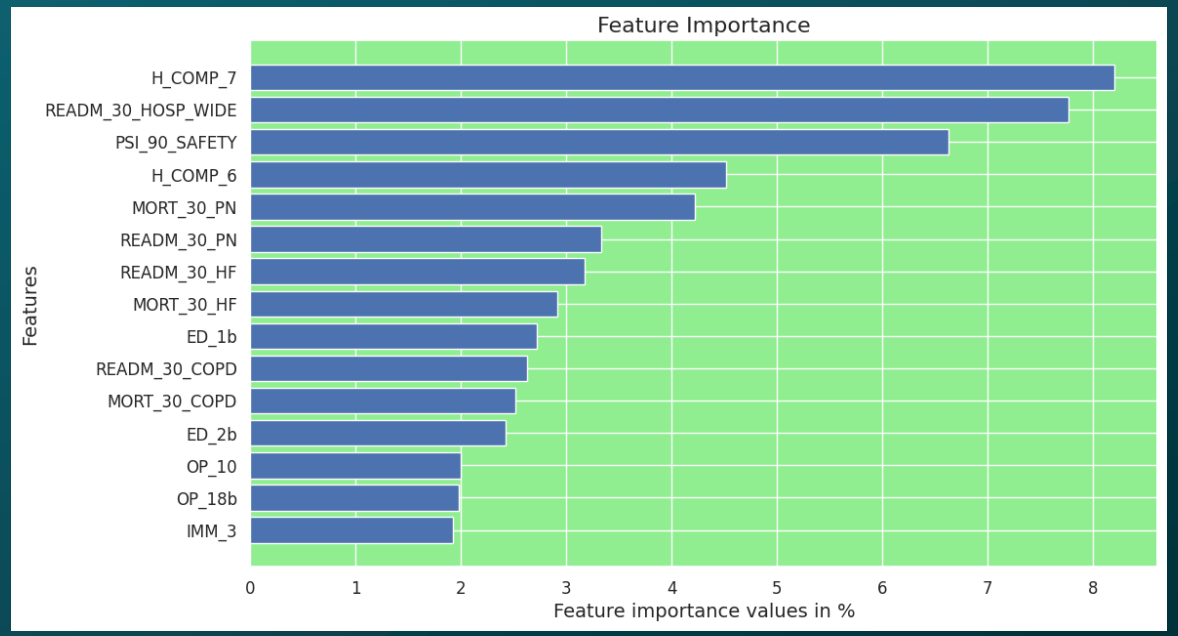
- **Reviewed CMS Hospital Compare methodology** to understand how 62 measures across 7 groups contribute to the overall Star Rating.
- **Selected 7 relevant datasets** from the provided flat files based on the measures linked to hospital performance and CMS rating criteria.
- **Performed data cleaning and exploratory analysis (EDA)** to understand data quality, distributions, correlations, and key problem areas.
- **Conducted univariate, bivariate, and multivariate analyses** to identify patterns and potential drivers of hospital ratings.
- **Prepared data for modeling**, including feature selection, encoding, handling missing values, and scaling where required.
- **Built and evaluated four predictive models**—Linear Regression, Logistic Regression, Decision Tree, and Random Forest—using hyperparameter tuning and stratified K-fold cross-validation.
- **Selected the best-performing model** based on accuracy and precision, then applied its best estimator specifically to Evanston Hospital.
- **Used the ELI5 library** to identify the top measures influencing Evanston's Star Rating and provide targeted improvement recommendations to help them move from a rating of **3 to at least 4**.

INSIGHTS, FINDINGS & VISUALIZATIONS

Star Rating Distribution Across Providers



Relative Feature Importance -



INSIGHTS, FINDINGS & VISUALIZATIONS

Linear Regression Model -

- Steps performed-
- Splitting the Data
- Feature Scaling
- Using RFE and checking VIF and p-value value.
- Residual Analysis
- Fitting in Test Data

The overall accuracy after doing Linear Regression was found out to be 59.5%.

Equation for best fitted line -

Rating = 3.090873 -0.357644 x PSI_90_SAFETY -0.054019 x OP_10 + 0.050121 x OP_13 + 0.058894 x OP_8 -0.053359 x MORT_30_AMI -0.119120 x MORT_30_COPD -0.074737 x MORT_30_HF -0.172764 x MORT_30_PN -0.043535 x MORT_30_STK + 0.048406 x READM_30_CABG + 0.132755 x READM_30_COPD + 0.071689 x READM_30_HF + 0.030086 x READM_30_HIP_KNEE -0.354792 x READM_30_HOSP_WIDE -0.174265 x ED_1b + 0.118215 x IMM_2 + 0.065893 x IMM_3 + 0.054150 x STK_1 + 0.082043 x VTE_1 + 0.112600 x H_COMP_7 -0.072443 x HAI_1

Logistic Regression

Steps followed:

- Splitting the data.
- Feature Scaling
- Applying Logistic Regression.
- Predicting the accuracy on Test Data.
- After performing Logistic Regression, we got an accuracy of 70%.

Random Forest

Steps included-

- Standard Scaling.
- Feature Importance.
- Hyper-parameter Tuning.
- Fitting the Final Model.
- Evaluation Matrix.
- PCA and Clustering.

MODEL PERFORMANCE SUMMARY

The performance of four models—Linear Regression, Logistic Regression, Random Forest, and XGBoost—was evaluated to predict hospital ratings.

Among these, **XGBoost delivered the highest accuracy of 81.4%**, outperforming all other models.

Both Random Forest and XGBoost showed strong training performance, but XGBoost maintained superior generalization on the test data.

Logistic Regression performed moderately, achieving around 70% accuracy.

Overall, **XGBoost emerged as the most reliable and accurate model**, making it the best choice for identifying key factors influencing Evanston Hospital's ratings.

Sr. No.	Type Of Model	Accuracy Score
1	Linear Regression	59.5%
2	Logistic Regression	70%
3	Random Forest	72%
4	XGBoost	81.4%

CONCLUSIONS & RECOMMENDATIONS

1) Safety of Care

Evanston Hospital performs below national standards in this category.

A major issue is **Catheter-Associated Urinary Tract Infections (CAUTI)**, often seen in bedridden patients.

Improving hygiene practices, ensuring proper sterilization, and strengthening staff accountability can significantly reduce these infections.

This is one of the easiest areas for the hospital to improve quickly.

2) Patient Experience

Cleanliness and hygiene around patient areas must be improved.

There is also a gap in effective doctor–patient communication.

Providing communication skill training for staff can enhance patient trust and satisfaction.

The hospital should review all patient interaction points, as experience ratings heavily influence overall scores.

3) Mortality Rates

The hospital has high mortality rates for several serious conditions:

30-day Heart Failure mortality

30-day Pneumonia mortality

30-day Acute Myocardial Infarction mortality

Improving clinical protocols and closely monitoring high-risk patients can help reduce these rates.

4) Readmission Rates

Overall readmission rates match national averages, but **COPD readmissions** remain high.

Stronger discharge planning, follow-up processes, and adherence to best recovery practices can reduce avoidable readmissions.

THANK YOU